



Great Learning

2-3 minutes

QUICK RECAP

In the previous week, we were introduced to Agentic AI and why AI agents are needed beyond traditional LLM-only approaches, learning how LLMs act as the reasoning core while agents perceive and act within different environments, and concluding with hands-on practice building simple single-agent systems to address business problems.

WEEK OVERVIEW

This week focuses on how AI agents go beyond LLMs by using external tools and memory to solve real-world business tasks. The sessions cover why tools are needed, key tool types (Python, SQL, Search, APIs), and how memory enables stateful, context-aware behavior through short-term and long-term retention. The week also introduces MCP as a standardized tool repository and concludes with hands-on practice building tool-based agents.

The following topics will be covered:

- Why Agents need external tools?
- Types of Tools (Python, SQL, Search, APIs)
- Why Memory matters? (stateless LLMs vs stateful agents)
- Short-term vs long-term memory
- Introduction to MCP (standardized tool repository)
- Tool-based Agents with MCP

LEARNING OBJECTIVES

By the end of this module, you will be able to:

- Explain why AI agents require external tools and how they extend the problem-solving capabilities of LLMs in real-world business contexts.
- Compare and assess different categories of tools (Python, SQL, Search, APIs) for their effectiveness in handling data-driven and operational tasks.
- Differentiate between stateless LLMs and stateful agents by analyzing the role of memory in enabling contextual continuity.
- Distinguish between short-term and long-term memory in agents and evaluate their business use cases (e.g., customer support vs. strategic planning).

Implement tool-based agents using MCP to standardize access to external tools and